

Multiple Choice (10 marks)

1. What is the most reasonable estimate of the length of a city's swimming pool?
 - (a) 1 meter
 - (b) 25 meters
 - (c) 1 kilometer
 - (d) 25 kilometers
2. The hypotenuse of a right triangle measures 10 inches and one angle is 45° . What is the number of square inches in the area of the triangle?
 - (a) 10
 - (b) 45
 - (c) 125
 - (d) 25
3. For the equation: $3x - 4(x - 2) + 6x - 8 = 0$, find the value of x .
 - (a) -4
 - (b) 0
 - (c) 2
 - (d) 4
4. In how many ways can 5 students be selected from a group of 6 students?
 - (a) 12
 - (b) 3
 - (c) 5
 - (d) 6
5. What is the domain of the function $f(x) = \frac{x+6}{\sqrt{x^2-3x-4}}$?
 - (a) $(-4, 4)$
 $0, 4$
 - (b) $(-\infty, -1) \cup (4, \infty)$
 - (c) $(-\infty, -1) \cup (-1, 4) \cup (4, \infty)$

True / False (10 marks)

Answer True or False

6. True or False: The greatest common factor of 42 and 84 is 21.
7. True or False: The smallest positive integer n such that $1/n$ is a terminating decimal and contains the digit 9 is 4096.
8. True or False: The value of y that satisfies the equation $y + 2.9 = 11$ is 9.1.
9. True or False: The value of N that satisfies the equation is 27, which is a common error when calculating exponent rules.
10. True or False: February has 28 days, which is 4 times the number of days in one week.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Find the ordered pair (a, b) of real numbers for which

$$(ax + b)(x^5 + 1) - (5x + 1)$$

is divisible by $x^2 + 1$.

12. Jimmy and Kima are planning a three-day trip where they will drive 182 miles, 439 miles, and 217 miles each day. Estimate the total distance they will drive and explain your reasoning.

End of Paper